STATION: SALEM 1 & 2

SYSTEM: Emergency Operating Procedures

TASK: Respond to a Loss of Heat Sink (Initiate Bleed and Feed with SI pumps and Rx Head Vents)

TASK NUMBER: N1150290501

JPM NUMBER: 13-01 NRC Sim d

ALTERNATE PATH: [X]

K/A NUMBER: EPE E05 EA1.1

IMPORTANCE FACTOR: 4.1 (**RO**) 4.0 (**SRO**)

APPLICABILITY:
EO [ ] **RO** [X] **STA** [ ] **SRO** [X]

EVALUATION SETTING/METHOD: Simulator - Perform

REFERENCES: 2-EOP-FRHS-1, Loss of Secondary Heat Sink, Rev. 24

TOOLS AND EQUIPMENT: None

VALIDATED JPM COMPLETION TIME: 6 minutes

TIME PERIOD IDENTIFIED FOR TIME CRITICAL STEPS: N/A

Developed By:	**G Gauding** **Instructor**	**Date: 9-6-14**
Validated By:	**C Omlor** **SME or Instructor**	**Date: 10-15-14**
Approved By:	Carney **Training Department**	**Date:** 10-23-14
Approved By:	Nicols **Operations Department**	**Date:** 10-27-14

ACTUAL JPM COMPLETION TIME:

ACTUAL TIME CRITICAL COMPLETION TIME:

PERFORMED BY:

GRADE: [ ] **SAT** [ ] **UNSAT**

REASON, IF UNSATISFACTORY:

EVALUATOR'S SIGNATURE: **DATE:**

NAME: ______________________

DATE: ______________________

SYSTEM: Emergency Operating Procedures

TASK: TCAF a Loss of Heat Sink (Initiate Bleed and Feed with SI pumps and Rx Head Vents)

TASK NUMBER: **N1150290501**

SIMULATOR SETUP: **RX HEAD VENT KEYS ARE LOCATED IN SIM BOOTH**

IC-254 developed by : MSL rupture (10%)downstream of MSIVs. Fail all MSIVs open. All AFW pumps failed. Fail **2PR2** shut. 21 CVCS pp C/T. Performed TRIP-1 through Step 20. 22 CVCS pp tripped during TRIP-1.

INITIAL CONDITIONS:

- Unit 2 initiated a Rx trip from 100% power in response to a Main Steamline break at the mixing bottle.
- An automatic Safety Injection initiated.
- The Main Turbine failed to trip automatically, and was manually tripped from the control console.
- MSLI failed, and all MSIV's remain open.
- All AFW flow has been lost.
- 21 charging pump is C/T.
- 22 charging pump tripped 3 minutes ago.
- EOP-TRIP-1 was performed and a transition to FRHS-1, Loss of Secondary Heat Sink was made at Step 20.

INITIATING CUE:

You are the Reactor Operator. Perform FRHS-1 starting at Step 1.

Successful Completion Criteria:
1. All critical steps completed.
2. All sequential steps completed in order.
3. All time-critical steps completed within allotted time.
4. JPM completed within validated time. Completion time may exceed the validated time if satisfactory progress is being made.

Task Standard for Successful Completion:
1. Stop ALL RCPs.
2. Open 2PR1.
3. Open Rx Head Vent Valves 2RC40-2RC43.

NAME: ____________________

DATE: ____________________

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TASK: **Respond to a Loss of Heat Sink (Initiate Bleed and Feed with SI pumps and Rx Head Vents)**

*	STEP NO.	STEP (* Denotes a Critical Step)	STANDARD	EVAL S/U	COMMENTS (Required for UNSAT evaluation)
			Operator states he has the watch.		
	1	IS TOTAL AFW FLOW LESS THAN 22E04 LB/HR DUE TO OPERATOR ACTIONS	Determines it was not operator action which caused total AFW flow to be less than 22E04 lb/hr.		
	2	IF AT LEAST ONE INTACT OR RUPTURED SG IS AVAILABLE, THEN DO NOT FEED A FAULTED SG	Recognizes ALL SGs are faulted.		
	3	IS RCS PRESSURE GREATER THAN ANY INTACT OR RUPTURED SG PRESSURE	Checks RCS pressure on control console and determines it is greater than all SG pressures checked on control console.		
	3.1	ARE RCS T-HOTS GREATER THAN 350°F	Checks RCS Thot indication on control console and determines that RCS Thots are greater than 350°F.		
	4	IS 21 OR 22 CHARGING PUMP AVAILABLE	Determines neither 21 nor 22 charging pump is available based on initial conditions and/or control console indications.		
		GO TO STEP 23	Goes to Step 23.		

NAME: ____________________

DATE: ____________________

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TASK: **Respond to a Loss of Heat Sink (Initiate Bleed and Feed with SI pumps and Rx Head Vents)**

*	STEP NO.	STEP (* Denotes a Critical Step)	STANDARD	EVAL S/U	COMMENTS (Required for UNSAT evaluation)
	23	CAUTION TO ESTABLISH RCS HEAT REMOVAL BY RCS BLEED AND FEED, STEPS 24 THRU 29 MUST BE PERFORMED QUICKLY AND WITHOUT INTERRUPTION	Reads Step.		
*	23	STOP ALL RCPS	Depresses STOP PB for 21-24 RCPs and verifies green stop light illuminates and red start light extinguishes.		
	24	INITIATE SI	Uses Safeguards key and initiates SI on at least one train of Safeguards initiation.		
	25	ARE SI VALVES IN SAFEGUARDS POSITION	Checks 2RP4 and/or console indication to determine that all valves listed in Table B are in Safeguards position. Table B valves are: 2SJ4 OPEN BIT INLET 2SJ5 OPEN BIT INLET 2SJ12 OPEN BIT OUTLET 2SJ13 OPEN BIT OUTLET 2CV68 CLOSED CHARGING DISCHARGE 2CV69 CLOSED CHARGING DISCHARGE (continued next page)		

NAME: ____________________

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SYSTEM: **Emergency Operating Procedures**

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*	STEP NO.	STEP (* Denotes a Critical Step)	STANDARD	EVAL S/U	COMMENTS (Required for UNSAT evaluation)
			Table B Valves (con't): 21-24SJ54 OPEN ACCUMULATOR OUTLET 2SJ1 OPEN RWST TO CHARGING 2SJ2 OPEN RWST TO CHARGING 2CV40 CLOSED DISCHARGE STOP 2CV41 CLOSED DISCHARGE STOP		
	25.1	IS 21 OR 22 CHARGING PUMP RUNNING	Checks control console and determines neither 21 nor 22 charging pump is running.		
	25.2	IS ANY SI PUMP RUNNING	Checks control console and determines both 21 and 22 SI pumps are running.		
	25.2	ARE VALVES IN TABLE C OPEN FOR AT LEAST ONE RUNNING SI PUMP	Checks control console indication for valves listed in Table C and determines the valves are open for at least one running SI pump. Table C valves are: 21/22 SI PUMPS 2SJ30 (FROM RWST) 21/22SJ33 (SI PUMP SUCTION) 2SJ135 (COLD LEG DISCHARGE) 21/22SJ134 (COLD LEG DISCHARGE)		
	26	OPEN BOTH PZR PORV STOP VALVES	Checks control console and determines BOTH 2PR6 and 2PR7 PORV STOP VALVES are open.		

OPERATOR TRAINING PROGRAM NAME: ____________

JOB PERFORMANCE MEASURE DATE: ____________

SYSTEM: **Emergency Operating Procedures**

TASK: **Respond to a Loss of Heat Sink (Initiate Bleed and Feed with SI pumps and Rx Head Vents)**

*	STEP NO.	STEP (* Denotes a Critical Step)	STANDARD	EVAL S/U	COMMENTS (Required for UNSAT evaluation)
*	26	OPEN BOTH PZR PORVS	Depresses 2PR1 PZR PORV open PB and verifies green closed light extinguishes and red open light illuminates. Depresses 2PR2 PZR PORV open PB and reports that 2PR2 will not open.		
	26.1	ARE BOTH PZR PORV STOP VALVES OPEN	Checks control console and determines BOTH 2PR6 and 2PR7 PORV STOP VALVES are open.		
		ARE BOTH PZR PORVS OPEN	Determines 2PR2 PZR PORV is not open.		
*	26.1	OPEN 2RC40 THRU 2RC43 (REACTOR HEAD VENTS)	Inserts key into each 2RC40 THRU 2RC43 (REACTOR HEAD VENTS) switch on 2RP3, turns to open, and verifies each valve opens.		
			Terminate JPM when operator has opened 2RC40 thru 2RC43 Reactor Head Vents.		

JOB PERFORMANCE MEASURE VALIDATION CHECKLIST

NOTE: All steps of this checklist should be performed upon initial validation. Prior to JPM usage, revalidate JPM using steps 8 and 11 below.

___ 1. Task description and number, JPM description and number are identified.

___ 2. Knowledge and Abilities (K/A) references are included.

___ 3. Performance location specified. (in-plant, control room, or simulator)

___ 4. Initial setup conditions are identified.

___ 5. Initiating and terminating Cues are properly identified.

___ 6. Task standards identified and verified by SME review.

___ 7. Critical steps meet the criteria for critical steps and are identified with an asterisk (*).

___ 8. Verify the procedure referenced by this JPM matches the most current revision of that procedure: Procedure Rev. 24 Date 10-15-14

___ 9. Pilot test the JPM:
   a. verify Cues both verbal and visual are free of conflict, and
   b. ensure performance time is accurate.

___ 10. If the JPM cannot be performed as written with proper responses, then revise the JPM.

___ 11. When JPM is revalidated, SME or Instructor sign and date JPM cover page.

SME/Instructor: [signature] Date: 10/15/14

SME/Instructor: ______________ Date: ______

SME/Instructor: ______________ Date: ______

INITIAL CONDITIONS:

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- An automatic Safety Injection initiated.
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INITIATING CUE:

You are the Reactor Operator. Perform FRHS-1 starting at Step 1.